AI for Applications

Matthew Renze
DATA SCIENCE CONSULTANT

@matthewrenze matthewrenze.com

Overview

AI Applications

Modular AI Applications

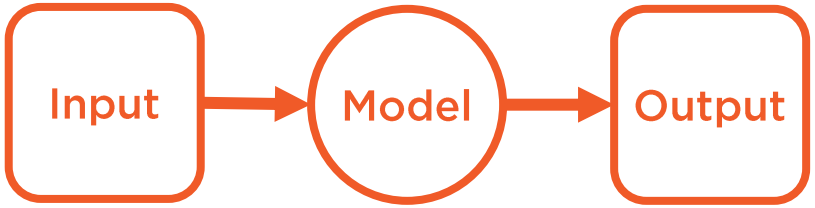
End-to-end AI Applications

Demo

Applications

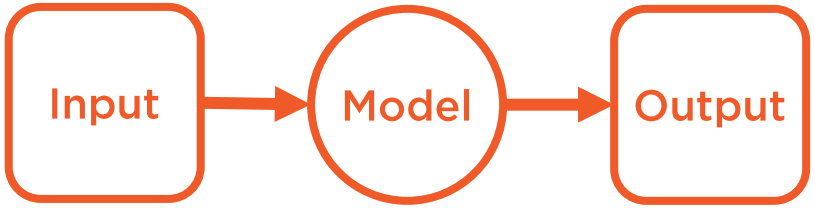
Applications

Simple AI Application

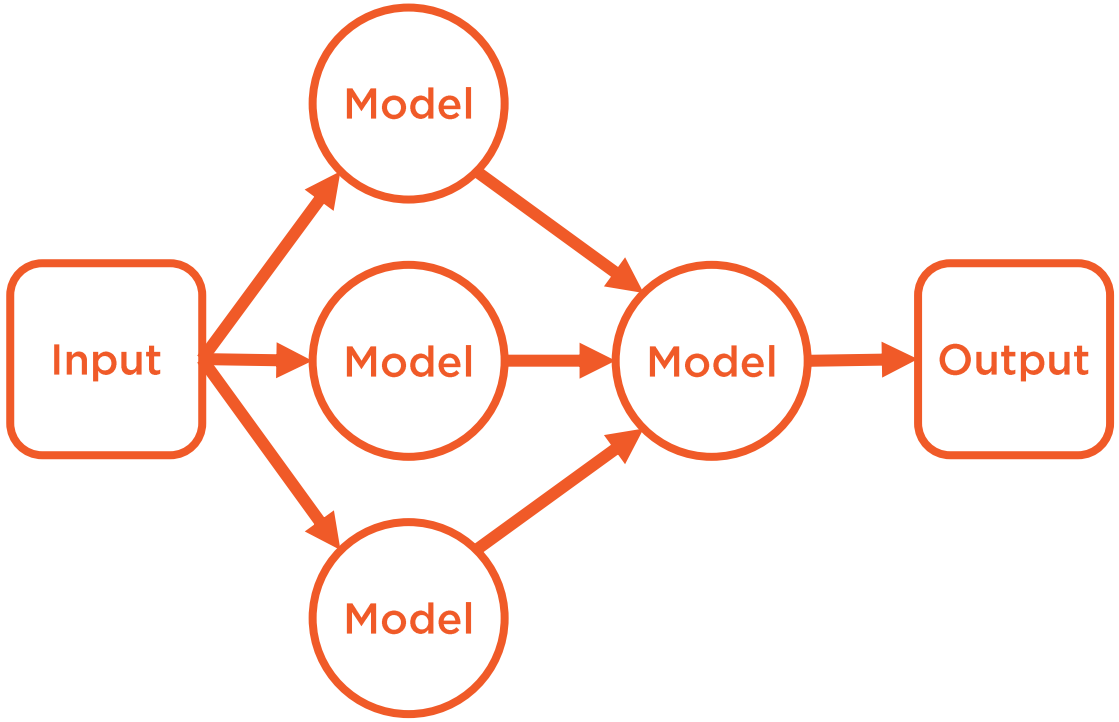


Applications

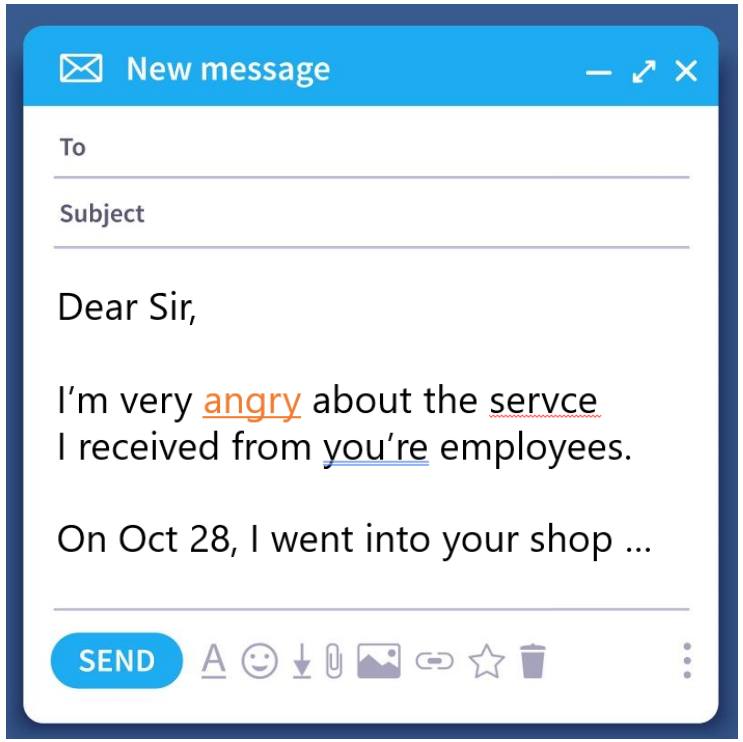
Simple AI Application

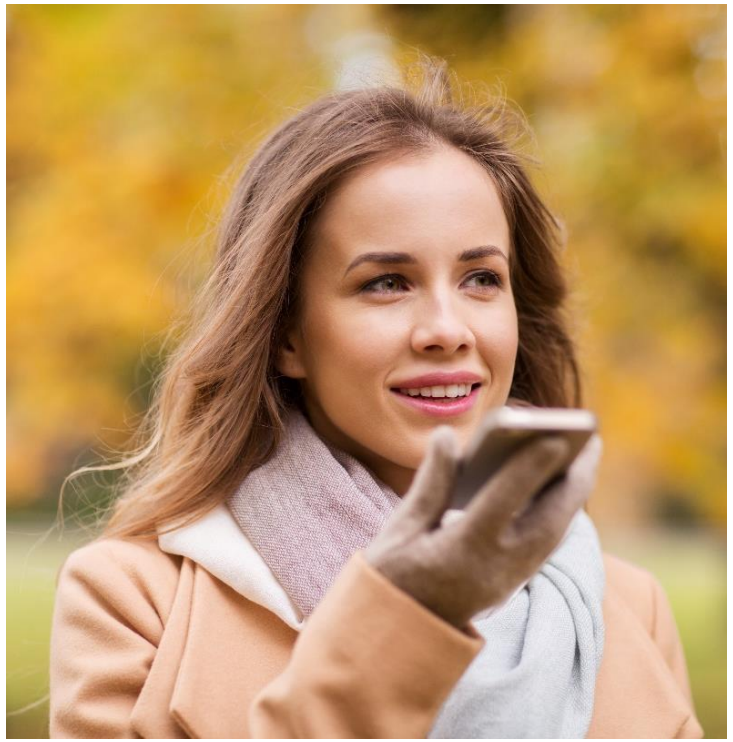


Complex AI Application



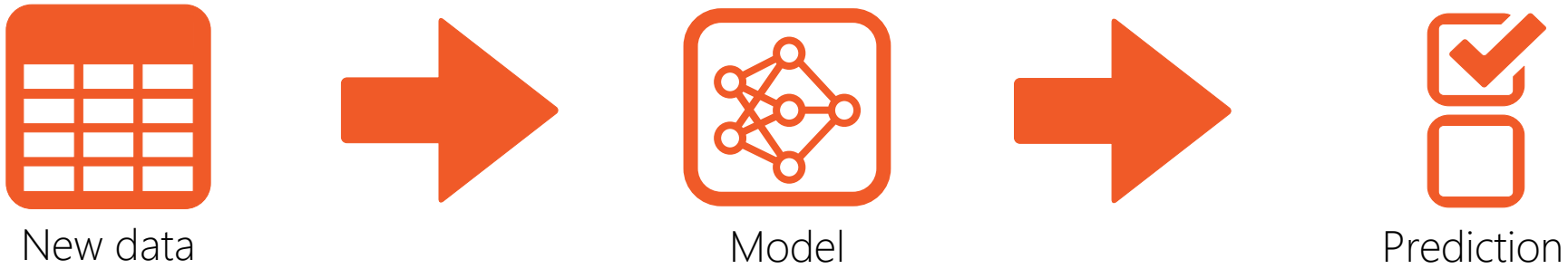
Applications





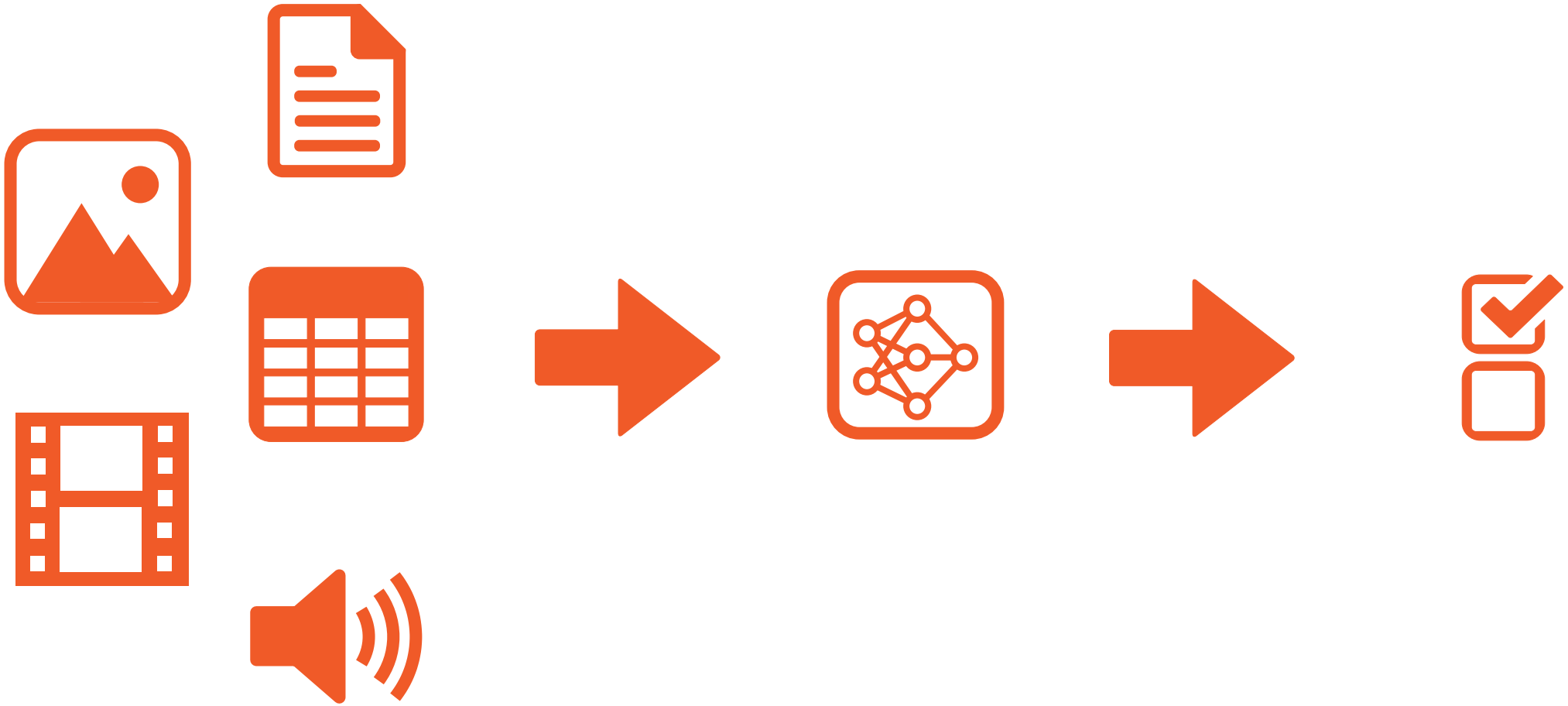
Modular AI Applications

It's All Just a Function

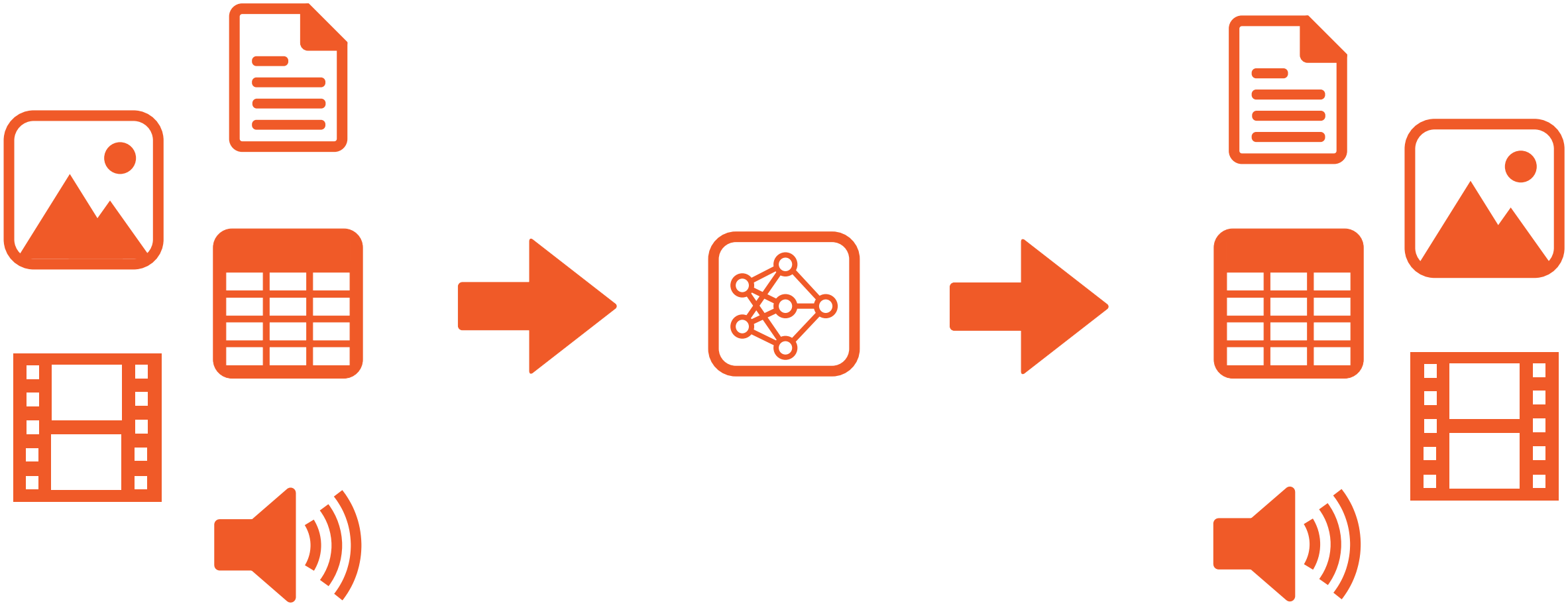


input $\longrightarrow$ $f(x)$ $\longrightarrow$ output

Analysis Functions

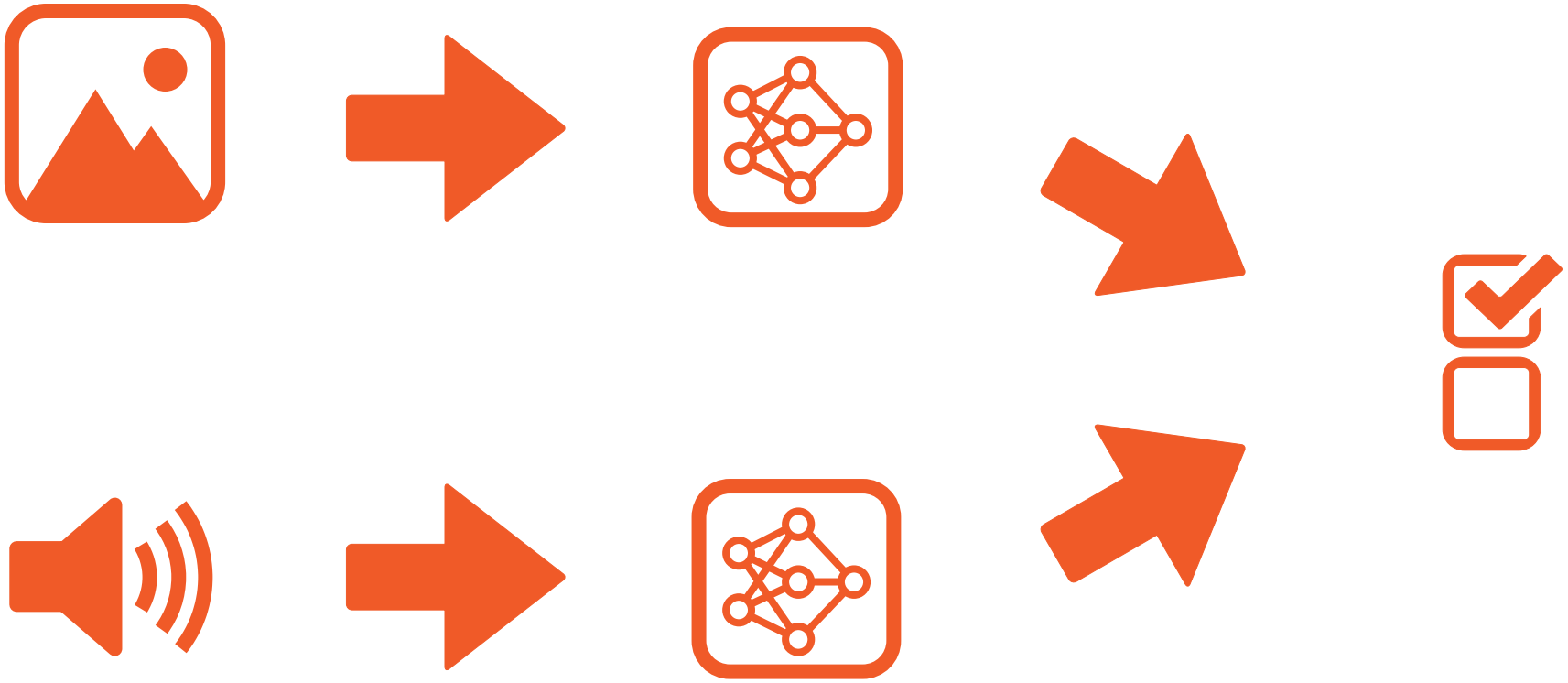


Synthesis Functions

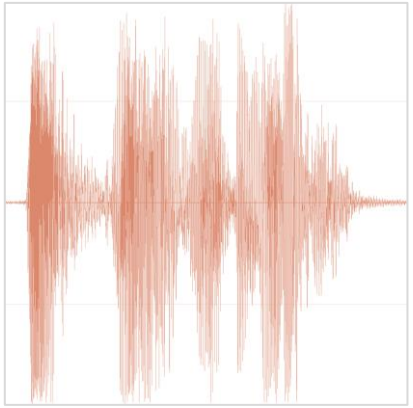


Other Functions

Combining Models



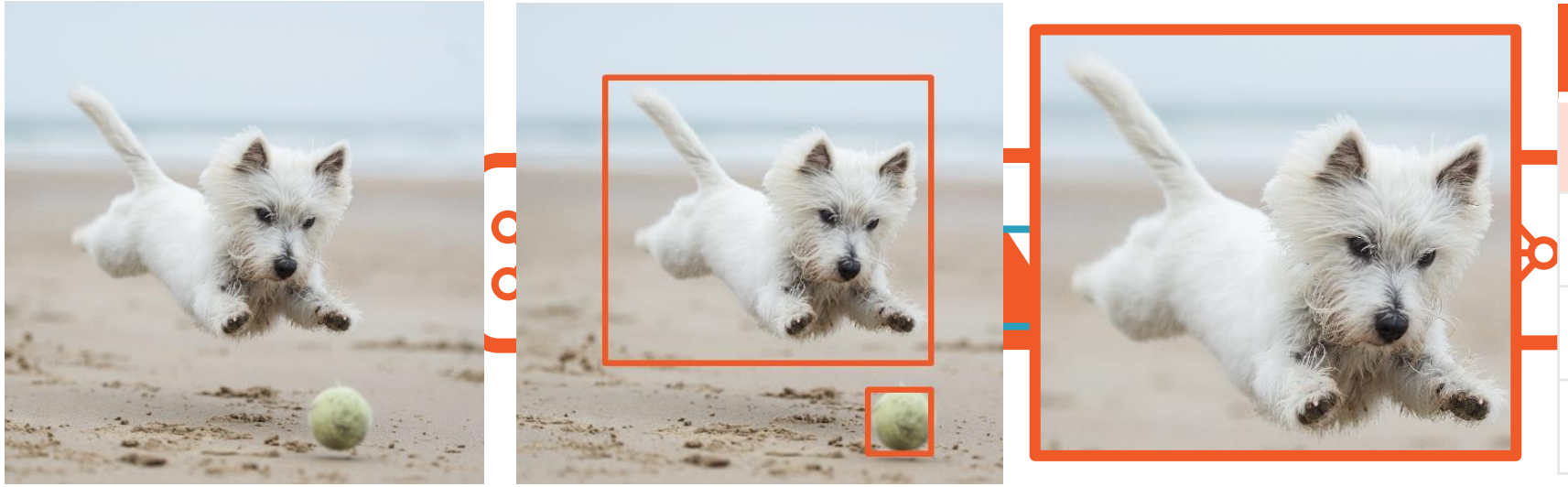
Combining Models



Output	
Class	Score
Matthew	99%
Zhang	50%
Priya	10%

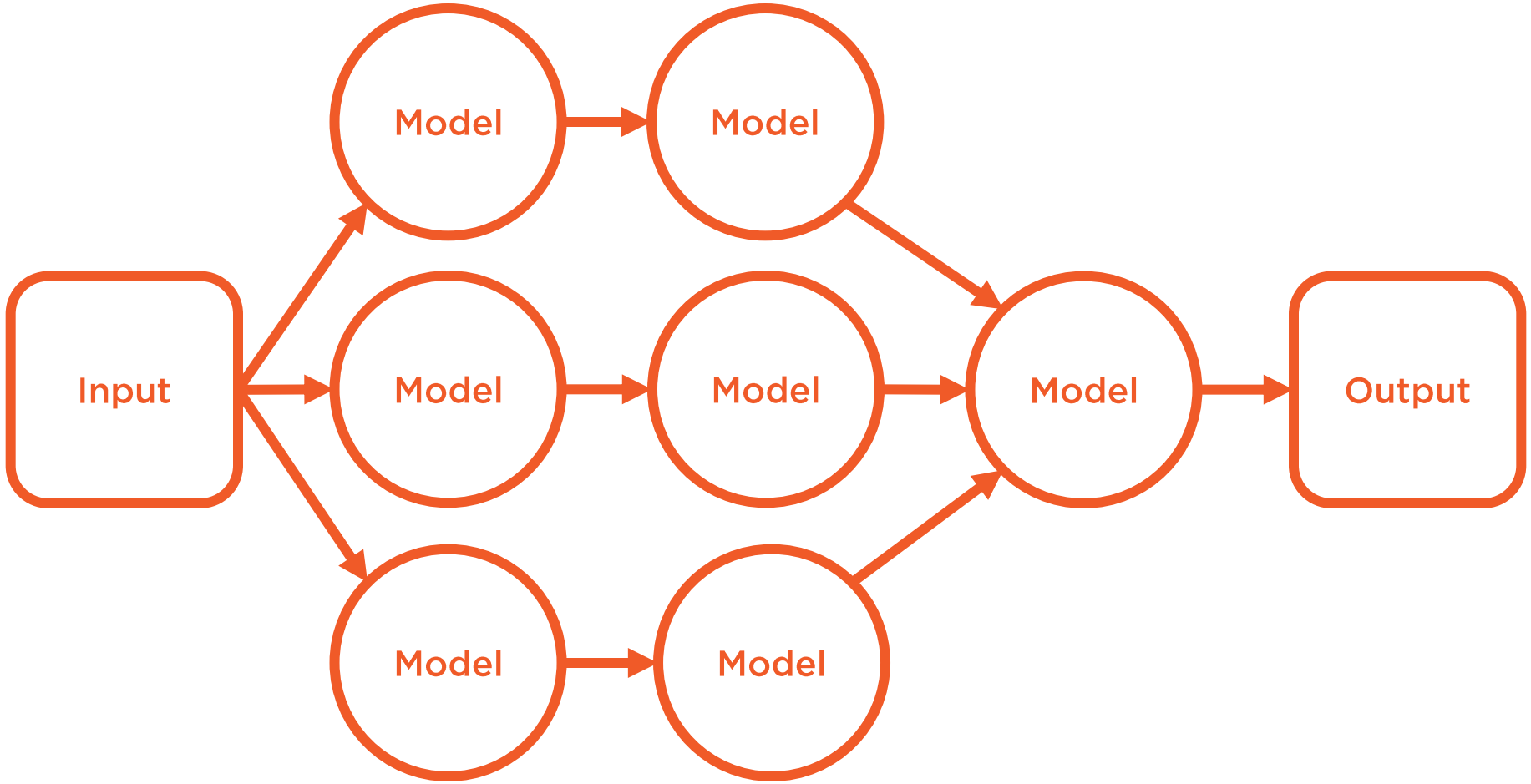
Chaining Models

Chaining Models

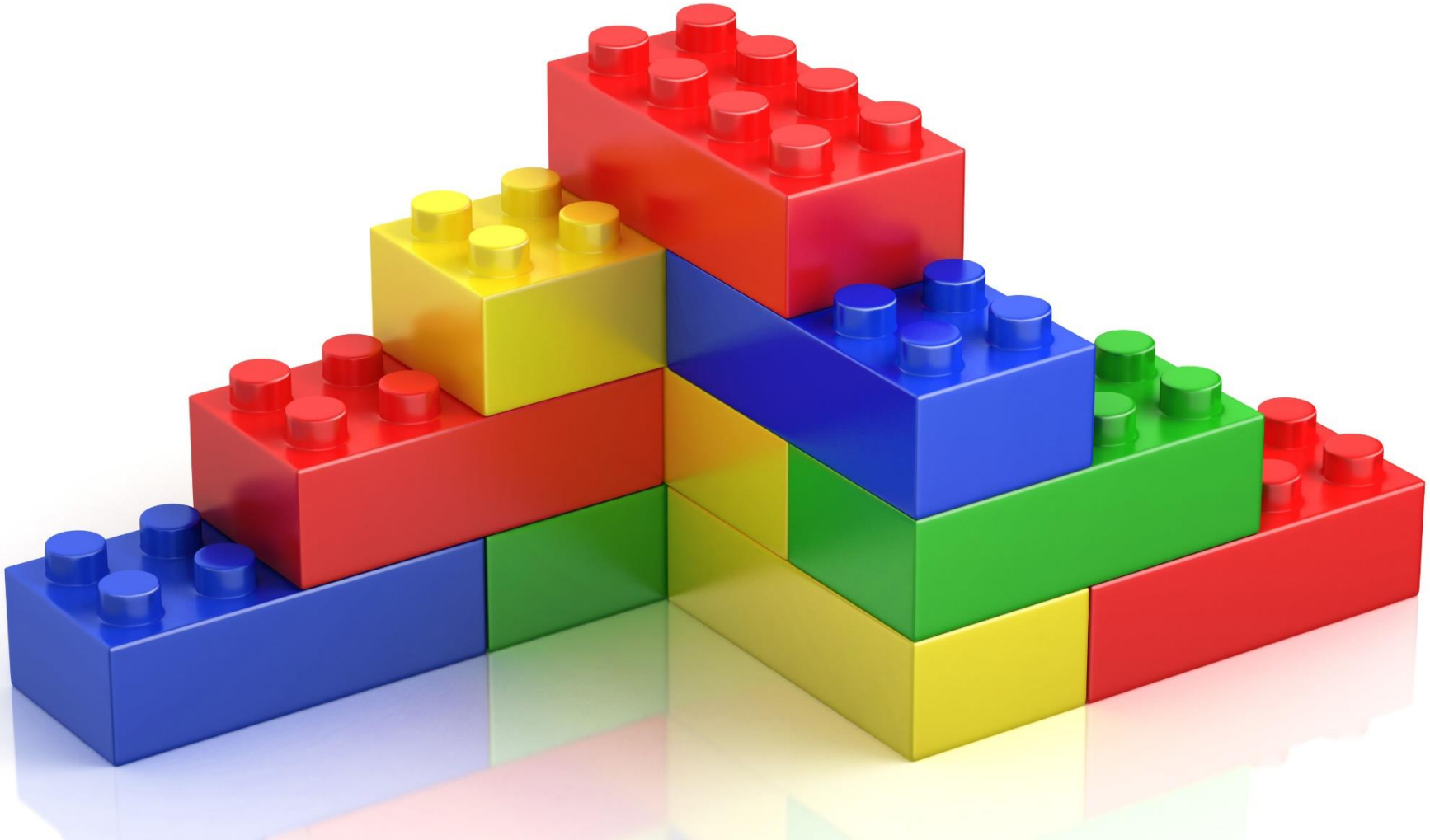


Output	
Class	Score
Dog	99%
Cat	50%
Mouse	10%

Modular AI Applications



It’s a Lot Like Legos!



End-to-end AI Applications

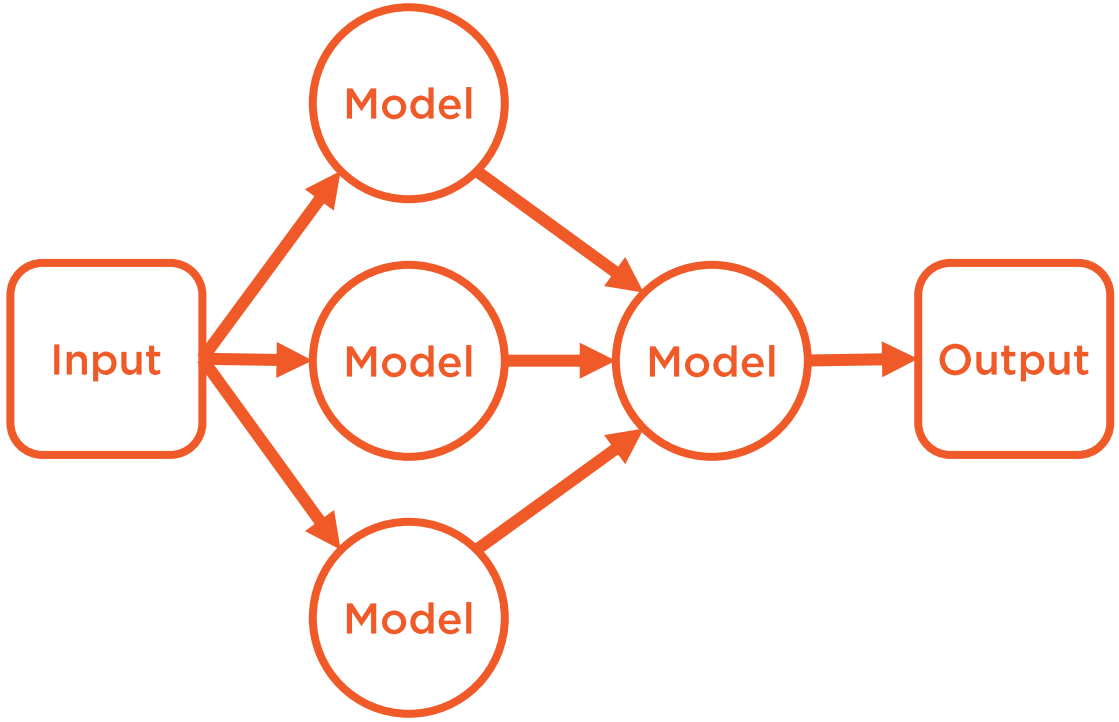
AI Application Types

Modular AI Application

End-to-end AI Application

AI Application Types

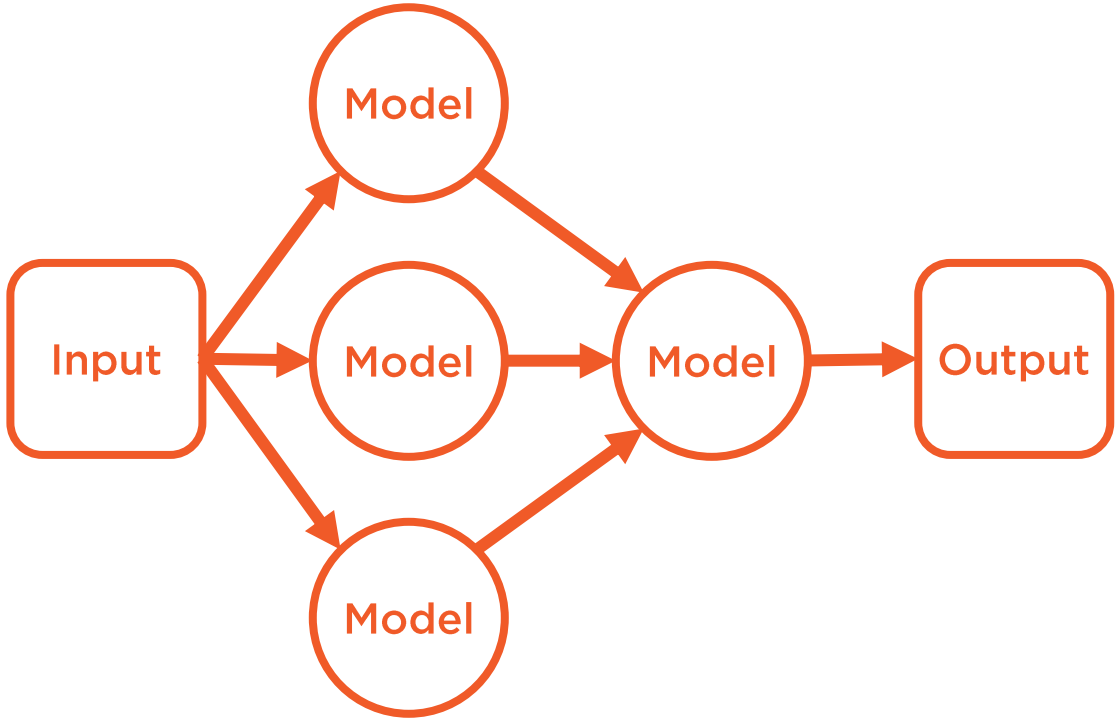
Modular AI Application



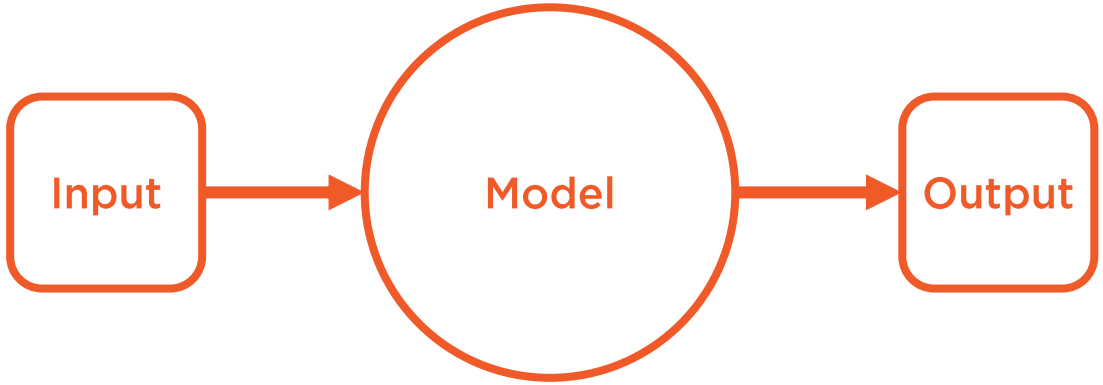
End-to-end AI Application

AI Application Types

Modular AI Application



End-to-end AI Application



AI Application Types

Modular AI Application

Pros

Cons

End-to-end AI Application

Pros

Cons

AI Application Types

Modular AI Application	End-to-end AI Application
Easier to create	
Easier to maintain	
Easier to debug	
Requires expertise	
Requires more code	

AI Application Types

Modular AI Application	End-to-end AI Application
Easier to create	More efficient
Easier to maintain	More powerful
Easier to debug	Less biased
Requires expertise	Harder to create
Requires more code	Not transparent

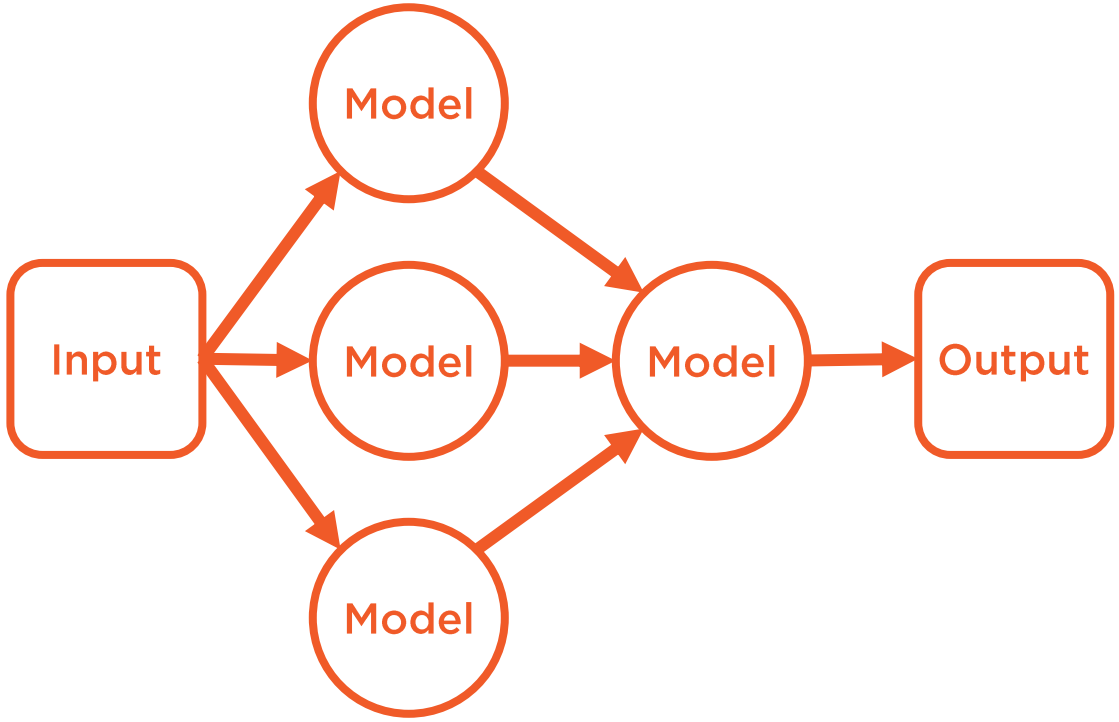
AI Application Types

Modular AI Application

End-to-end AI Application

AI Application Types

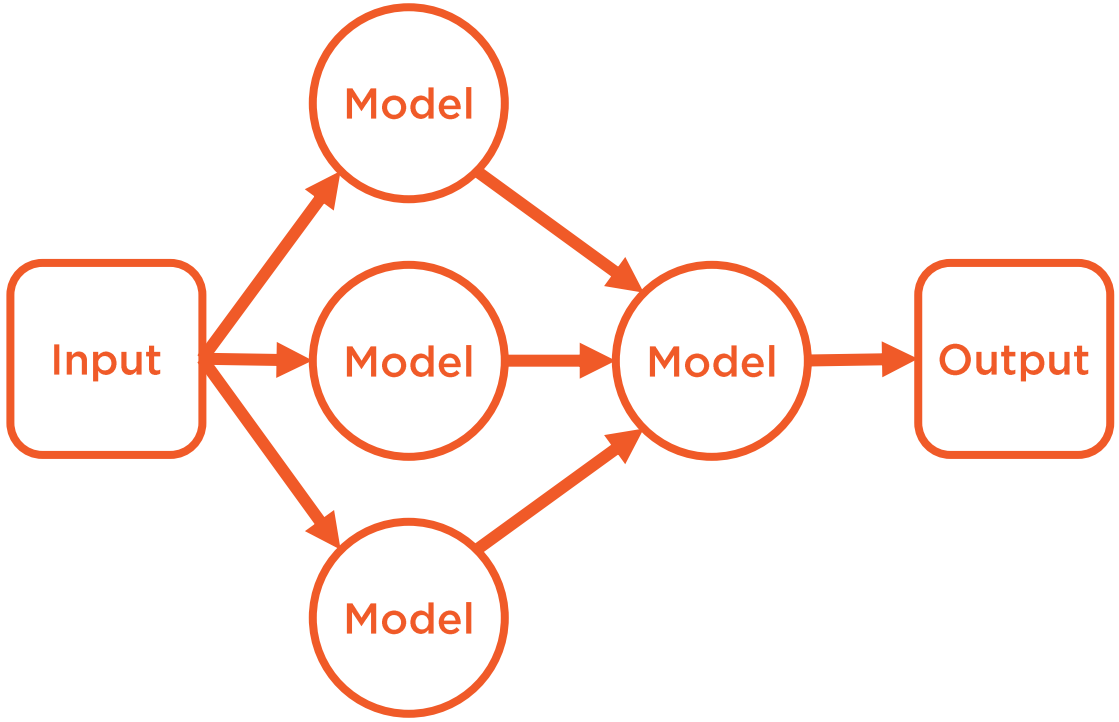
Modular AI Application



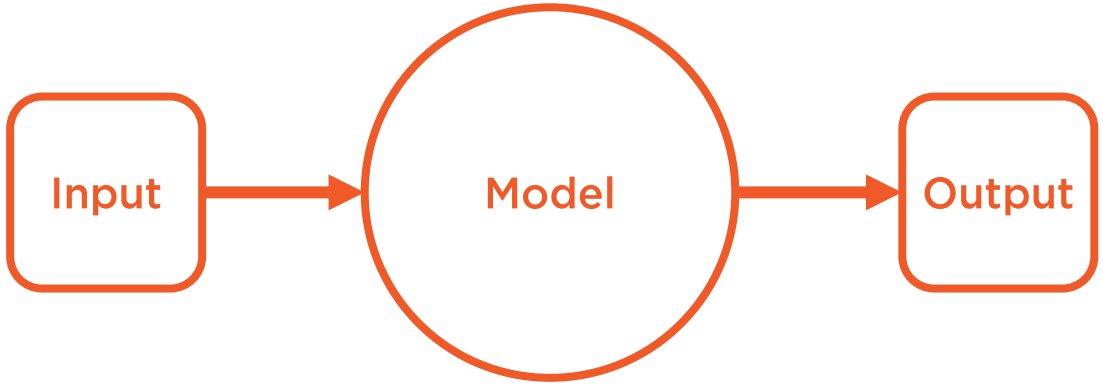
End-to-end AI Application

AI Application Types

Modular AI Application

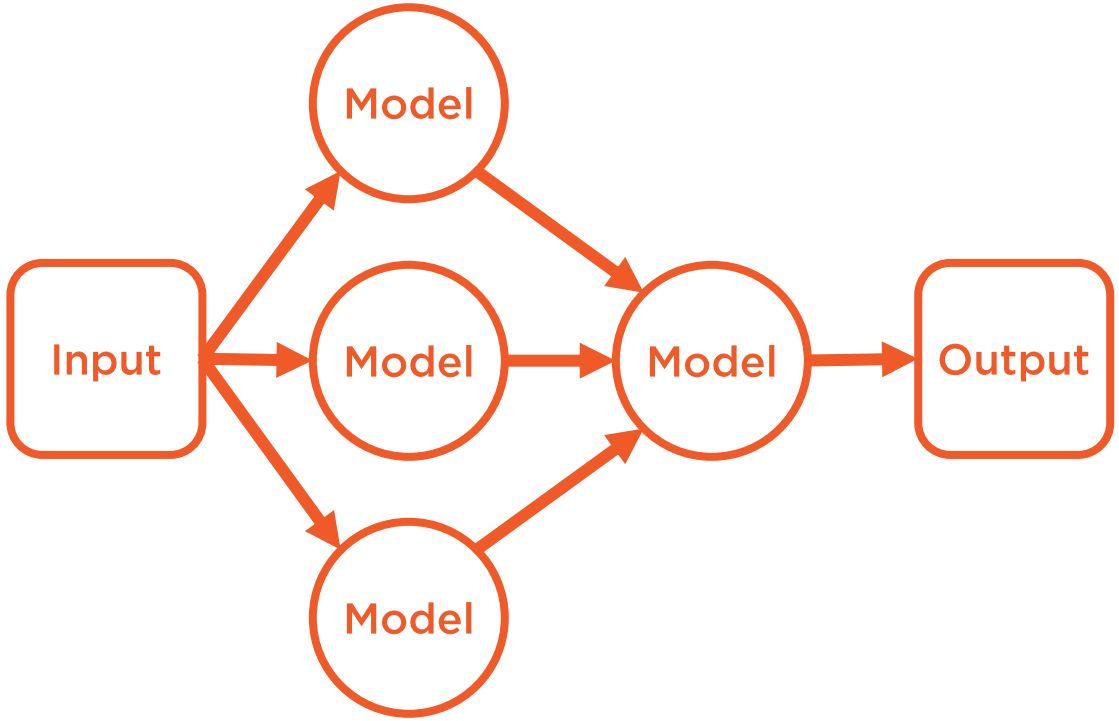


End-to-end AI Application

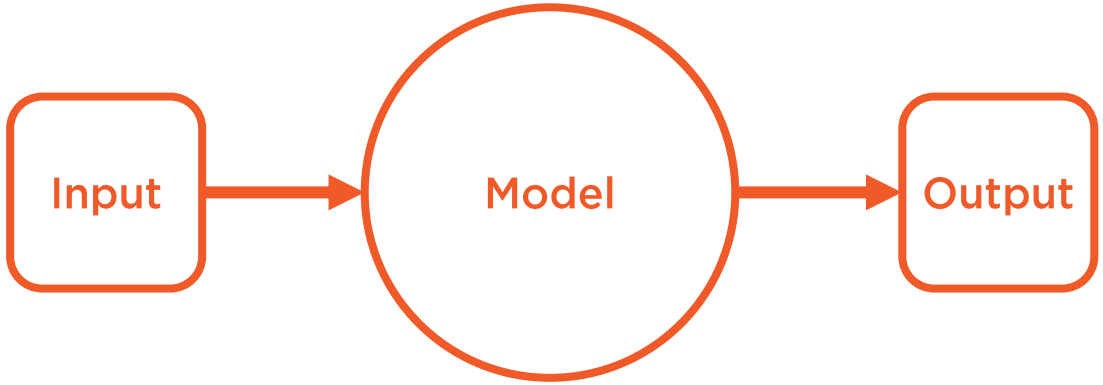


AI Application Types

Modular AI Application



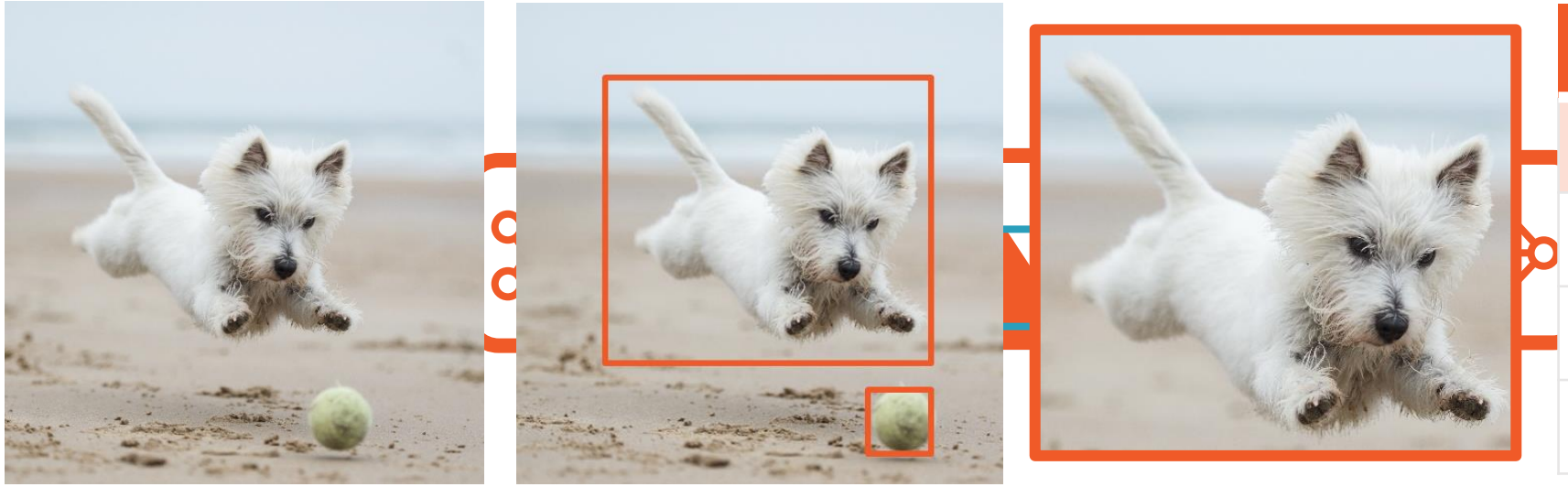
End-to-end AI Application



Easy → Hard

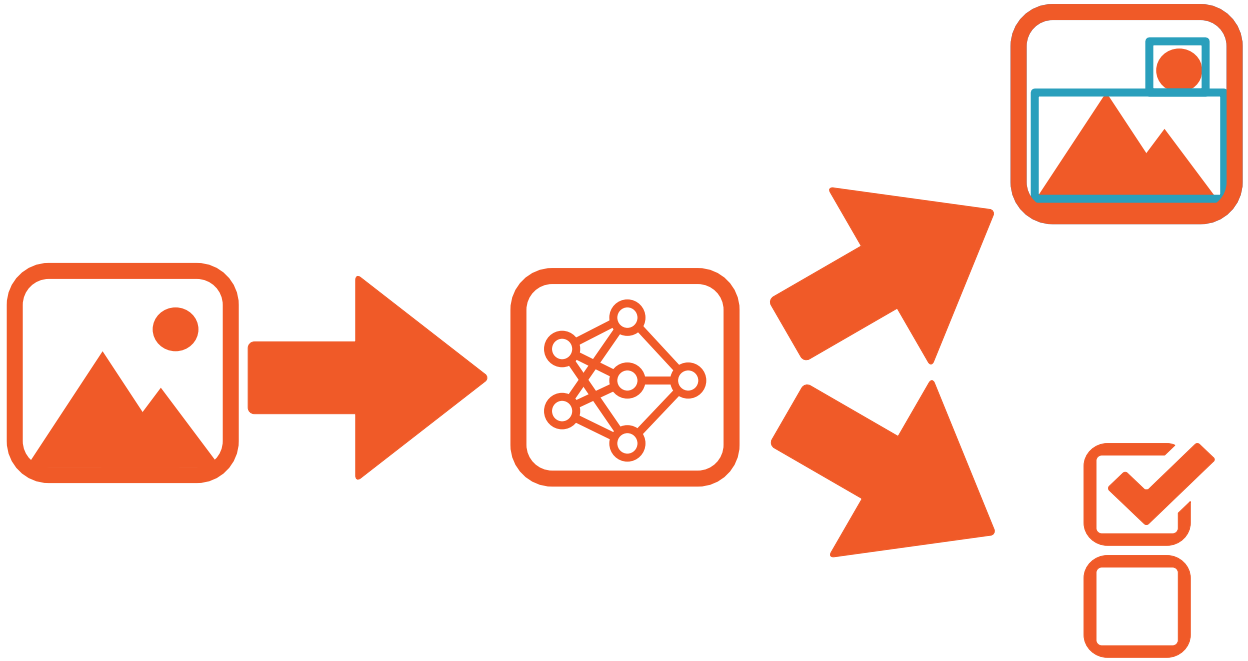
Demo

Demo Overview



Output	
Class	Score
Terrier	77%
Dog	82%
Animal	88%

Vision Services



Demo Options

Microsoft

REST API

Computer-Vision

Cognitive Services

Search (Ctrl+/)

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems

RESOURCE MANAGEMENT

- Quick start
- Keys and Endpoint
- Pricing tier
- Networking
- Identity
- Billing By Subscription
- Properties
- Locks

Monitoring

- Alerts
- Metrics

Delete

Essentials

JSON View

Resource group (change)
Cognitive-Services

Status
Active

Location
East US

Subscription (change)
RenzeConsulting

Subscription ID
[redacted]

Tags (change)
Click here to add tags

API type
Computer Vision

Pricing tier
Free

Endpoint
https://computer-vision.cognitiveservices.azure.com/

Manage keys
Click here to manage keys

Quota info

Free tier Total
5000 Calls

Free tier Remaining
5000 Calls

Free tier Rest period
30.00 Days

https://github.com/matthewrenze/ai-developers-toolkit/

Summary

Applications

Modular Applications

End-to-end Applications

Demo